

blinkit Power BI Dashboard Project

Project Statement

Context:

Blinkit, a quick-commerce platform, enables users to order groceries and essentials for delivery in minutes. For businesses operating at this scale, effective data management and visualization are crucial for monitoring performance, identifying trends, and optimizing operations.

Challenge:

Managing and analyzing a large volume of data related to orders, inventory, delivery times, customer feedback, and operational metrics can be overwhelming. Without a centralized dashboard, decision-making becomes inefficient, impacting overall performance and customer satisfaction.

Goals:

1. Develop a comprehensive dashboard to consolidate and visualize key performance indicators (KPIs) such as:
 - Number of daily/weekly/monthly orders.
 - Delivery time trends and bottlenecks.
 - Inventory status and restocking requirements.
 - Customer satisfaction ratings and feedback analysis.
 - Revenue and profit margins.
2. Enable real-time updates and dynamic filtering for interactive data exploration.
3. Provide actionable insights to improve operational efficiency and customer experience.

Target Audience:

- Business stakeholders looking for high-level summaries of performance.
- Operations managers requiring granular data for process optimization.
- Data analysts and engineers exploring the backend data for advanced insights.

Key Features:

1. Intuitive charts and graphs (e.g., bar charts, line graphs, and pie charts) for quick understanding.
2. Drill-down capabilities for detailed data exploration.
3. Filters to segment data by time periods, regions, or product categories.
4. Export options to share reports with team members.

Expected Outcomes:

- A user-friendly dashboard that empowers stakeholders to make data-driven decisions.
- Increased operational efficiency through enhanced data visibility.
- A template or framework that can be adapted to similar quick-commerce businesses.

Why This Project?

The Blinkit Dashboard Project is an excellent opportunity to learn about:

- Advanced data visualization techniques using Power BI.
- Integrating real-world datasets for meaningful analysis.
- Solving business challenges through technology and analytics.

Project Components

Sales Overview Page -:

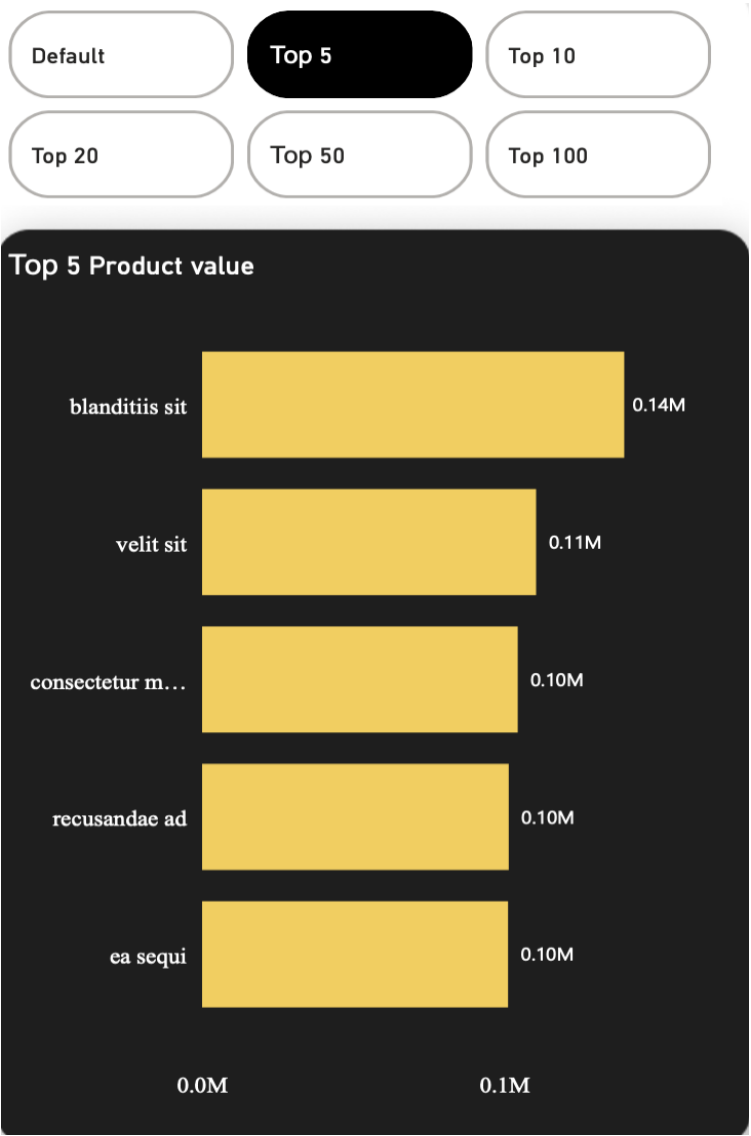
1. Revenue by Product Category (Column Chart)

Data Shown:

- This bar chart highlights the top products contributing the highest total value in sales.
- The user interface also features slicer options (Top 5, 10, 20, 50, 100), allowing dynamic comparison across different product tiers.

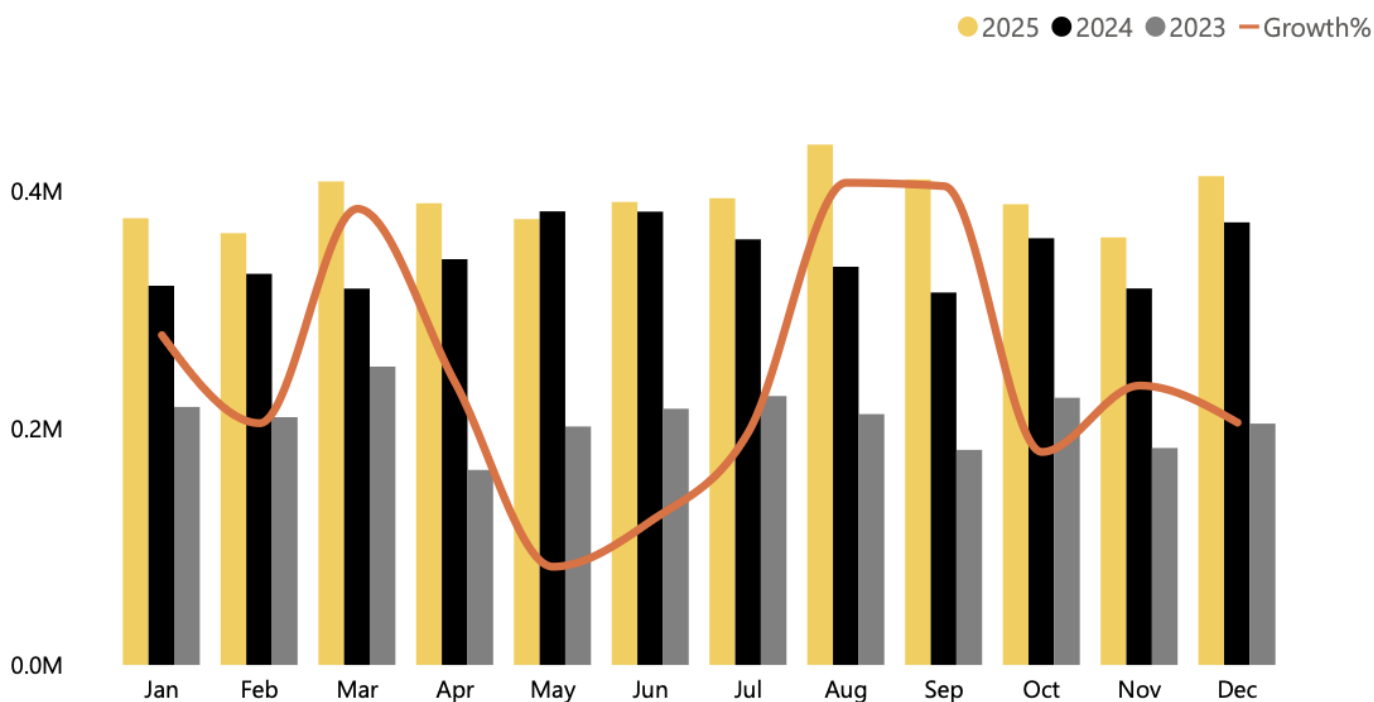
Business Implication:

- Identifying the top-performing products helps Blinkit focus on inventory planning, demand forecasting, and targeted marketing.
- These top 5 products are the key revenue drivers; hence, the business should ensure consistent stock availability and consider bundling or promotional strategies around them.
- Insights from this chart can also guide supplier negotiation, optimize warehouse space, and drive up-sell strategies on the app.
- The slicer feature enhances decision-making by allowing stakeholders to quickly analyze performance beyond the top 5, improving category-level insights.



2. Monthly Sales Trends (Line Chart)

2025, 2024, 2023 and Growth% by Month



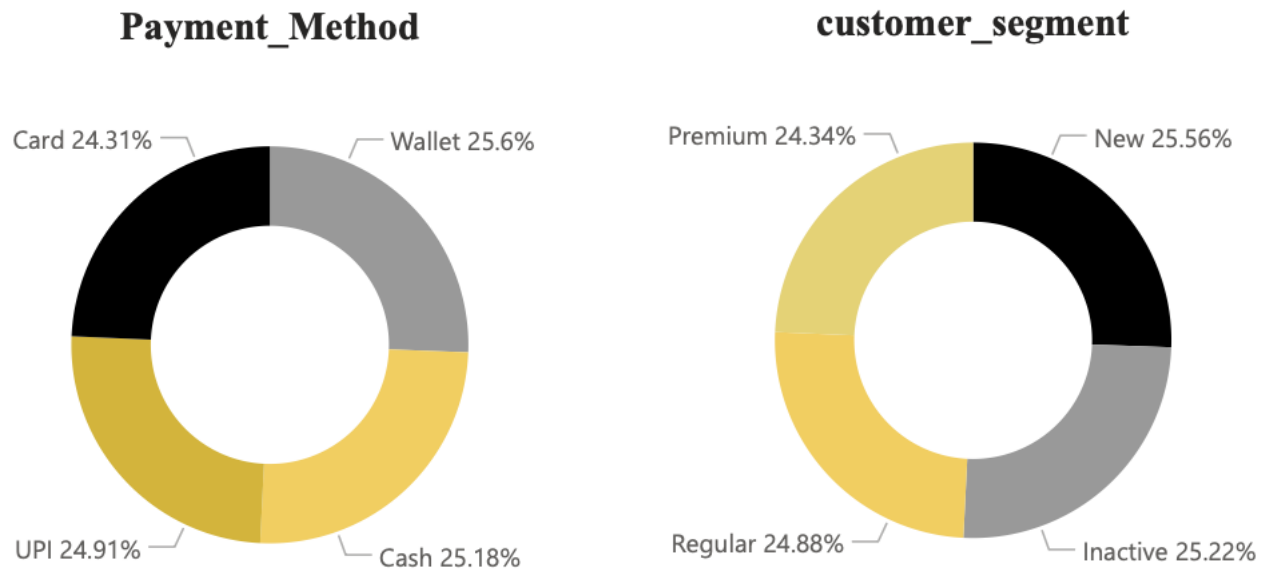
Data Shown:

- This multi-bar and line chart presents monthly sales performance over three years (2023–2025) alongside the **Growth% trend**.
- The bars represent monthly sales for each year (gray for 2023, black for 2024, gold for 2025), while the orange line indicates the year-over-year growth percentage.
- A noticeable upward trajectory is observed in 2025 across most months compared to previous years, especially in **March, August, and September**, where sales values are significantly higher.
- The **Growth%** line shows fluctuations, with peaks in **March and August** and dips around **May and October**, indicating seasonal or operational impact.

Business Implication:

- The chart reveals strong **YoY growth**, suggesting effective strategies or market expansion efforts contributing to rising sales.
- Peak months like **August and March** indicate seasonal demand or successful campaigns — these should be further analyzed and reinforced in planning.
- Months showing lower growth (like **May and October**) highlight potential areas for improvement in customer engagement, supply chain management, or promotions.
- Tracking growth over multiple years allows for more accurate **forecasting, budget planning, and resource allocation** in upcoming periods.
- The inclusion of Growth% also helps stakeholders assess not just volume but efficiency and momentum of growth.

4. Payment Method and Customer Segment Distribution (Donut Chart) -:



Visualization 1: Payment Method Distribution

Data Shows:

- The donut chart displays the distribution of payment methods used by customers on Blinkit.
- **Wallet (25.6%)** is the most used method, followed by **Cash (25.18%)**, **UPI (24.91%)**, and **Card (24.31%)**.
- All four methods are used almost equally, indicating diversified customer preferences.

Business Implication:

- Since Wallet and UPI account for over 50% of total transactions, Blinkit should continue strengthening its support and offers around digital wallets and UPI systems.
- The near-equal spread implies the need to maintain seamless experiences across all payment channels, ensuring no friction.
- Blinkit can target specific payment-based promotions (like cashback on UPI or Wallet) to further increase retention and repeat purchases.

Visualization 2: Customer Segment Distribution

What the Data Shows:

- The donut chart presents the share of different customer segments:
 - **New (25.56%)**
 - **Inactive (25.22%)**
 - **Regular (24.88%)**
 - **Premium (24.34%)**
- The distribution is quite balanced, but nearly **50% of customers** fall under New and Inactive categories.

Business Implication:

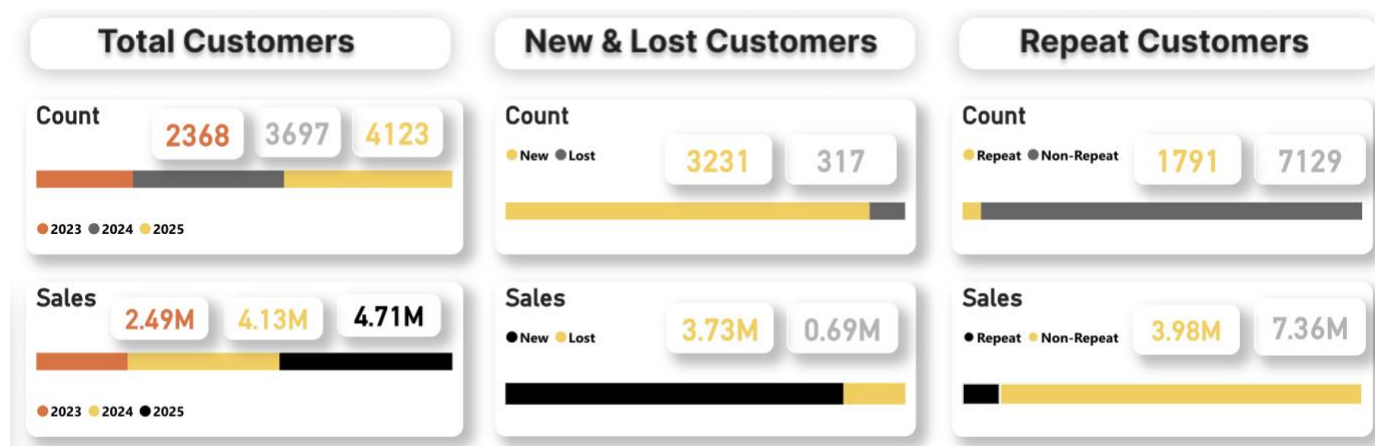
- A large proportion of **New** users implies strong marketing or referral acquisition, which is positive.
- However, the high percentage of **Inactive** customers suggests a drop-off after initial engagement.

- Strategies like retargeting, loyalty programs, or personalized offers can help convert Inactive and New users into Regular or Premium customers.
- Improving app experience, delivery reliability, and value-added services may reduce churn and boost customer lifetime value.

Customer Page -:

1. Customer Analysis — Total, New, Lost & Repeat (2023–2025) (Cards)

Data Shows:



1. Total Customers

- **Customer count** has steadily grown over the years:
 - 2023: **2,368**
 - 2024: **3,697**
 - 2025: **4,123**
- Corresponding **sales also increased**:
 - 2023: **2.49M**
 - 2024: **4.13M**
 - 2025: **4.71M**

2. New & Lost Customers

- **New Customers:** 3,231
- **Lost Customers:** 317
- Net gain shows healthy acquisition with low churn.
- Sales from new customers: **3.73M**, vs. only **0.69M** from lost ones, highlighting the importance of acquisition.

3. Repeat vs Non-Repeat Customers

- **Repeat Customers:** 1,791 | Sales: **3.98M**
- **Non-Repeat Customers:** 7,129 | Sales: **7.36M**
- Though non-repeat customers dominate in count, repeat customers offer higher value per customer, indicating strong brand loyalty and retention ROI.

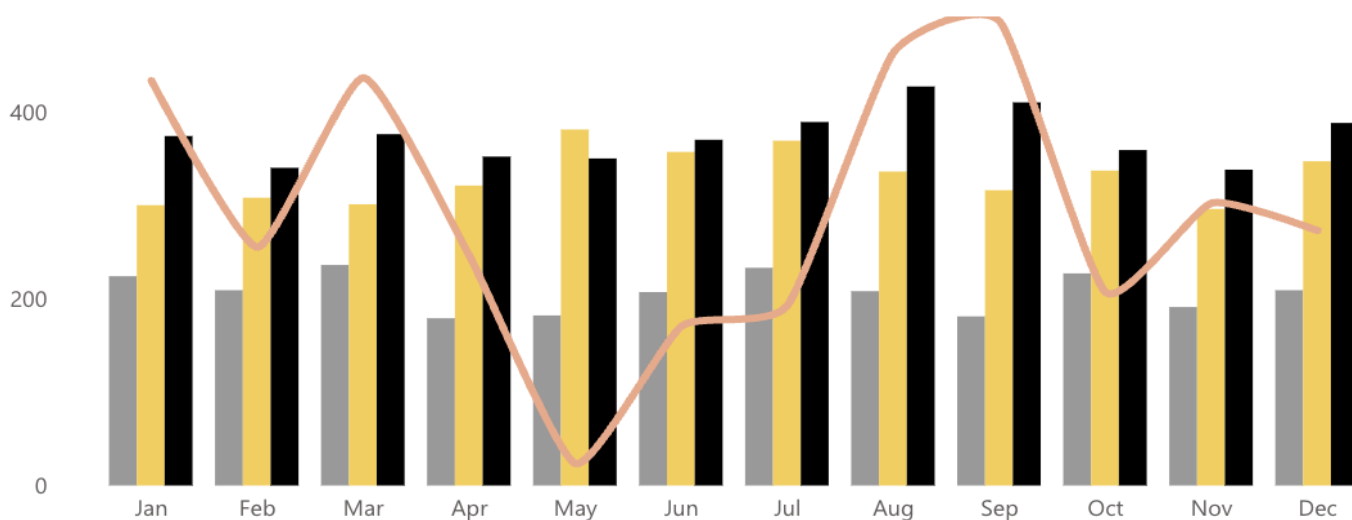
Business Implication:

- Consistent year-over-year customer growth demonstrates effective **marketing and customer experience strategies**.
- High volume of **new customers** and minimal losses in 2025 reflect robust **customer acquisition and retention** efforts.
- The strong contribution of **repeat customers to revenue** underscores the importance of nurturing existing relationships through loyalty programs or personalized offers.
- While non-repeat customers generate higher total sales due to volume, converting a portion of them into repeat customers could significantly boost long-term profitability.

2. Monthly Customer Counts and Growth Trend (2023–2025)

Customer Counts and Growth By Month

Year ● 2023 ● 2024 ● 2025 ● Customer_Growth



Data Shows:

- Year-over-Year Growth:**
 - There is a clear **upward trend** in customer counts from 2023 to 2025 across most months.
 - The black bars (2025) consistently lead in height, indicating significant **customer acquisition improvements** over the years.
- 2025 Monthly Growth Fluctuations:**
 - The **Customer Growth** line (peach-colored) shows **peaks in March, August, and September**, indicating strong acquisition during these months.
 - **Growth dips in May and October**, suggesting a slowdown in customer onboarding or seasonal inactivity.
- Consistent Performance:**
 - Despite fluctuations in growth, 2025 maintains **high and stable customer counts**, especially from **June to December**, reflecting a solid retention strategy.

Business Implication:

- Marketing and outreach campaigns in **March, August, and September** seem highly effective and could be analyzed for best practices.
- **Growth dips in May and October** may need further investigation to address potential bottlenecks or seasonality impacts.
- The upward trajectory year-over-year supports a **positive long-term customer acquisition strategy**, reinforcing the effectiveness of business development initiatives.

3. Feedback Category & Sentiment Distribution (Donut Chart)

Chart 1: Feedback Category Distribution

This chart illustrates the proportion of customer feedback categorized into four key areas:

- **Customer Service:** 25.34%
- **Product Quality:** 25.11%
- **Delivery:** 24.94%
- **App Experience:** 24.61%

Data Shows:

- The **distribution is fairly balanced**, indicating that customers are giving feedback across all major operational touchpoints.
- **Customer Service** holds the highest share, closely followed by **Product Quality** and **Delivery**, implying these are top areas of concern or interest for users.

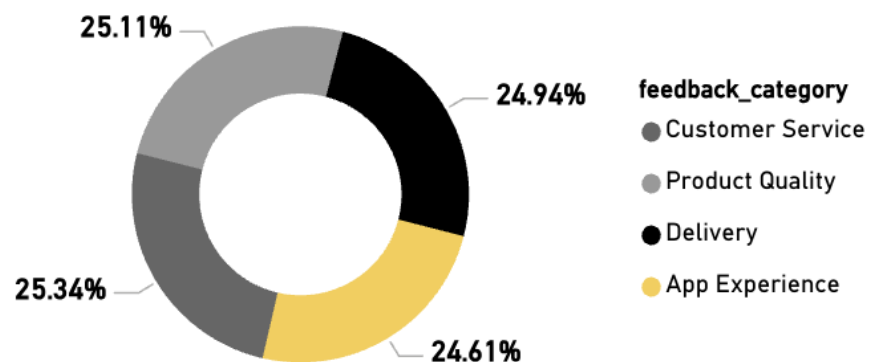


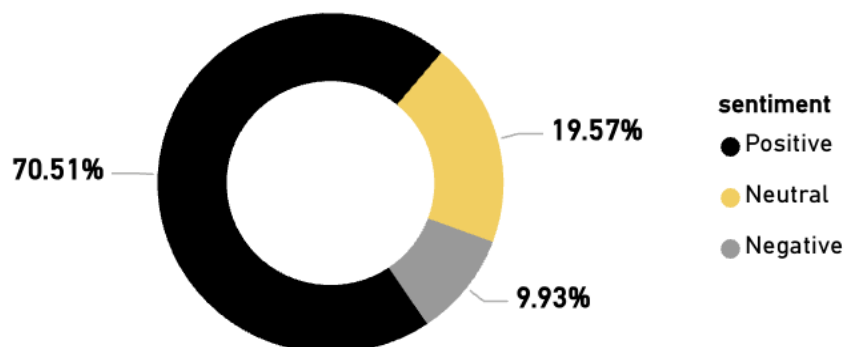
Chart 2: Sentiment Distribution

This donut chart classifies customer sentiments as follows:

- **Positive:** 70.51%
- **Neutral:** 19.57%
- **Negative:** 9.93%

Data Shows:

- A strong **majority of feedback is positive** (over 70%), which is a great indicator of overall customer satisfaction.
- **Negative sentiment is below 10%**, suggesting that pain points exist but aren't widespread.
- **Neutral feedback** (almost 20%) may represent areas where expectations were met but not exceeded potential room for delighting customers.

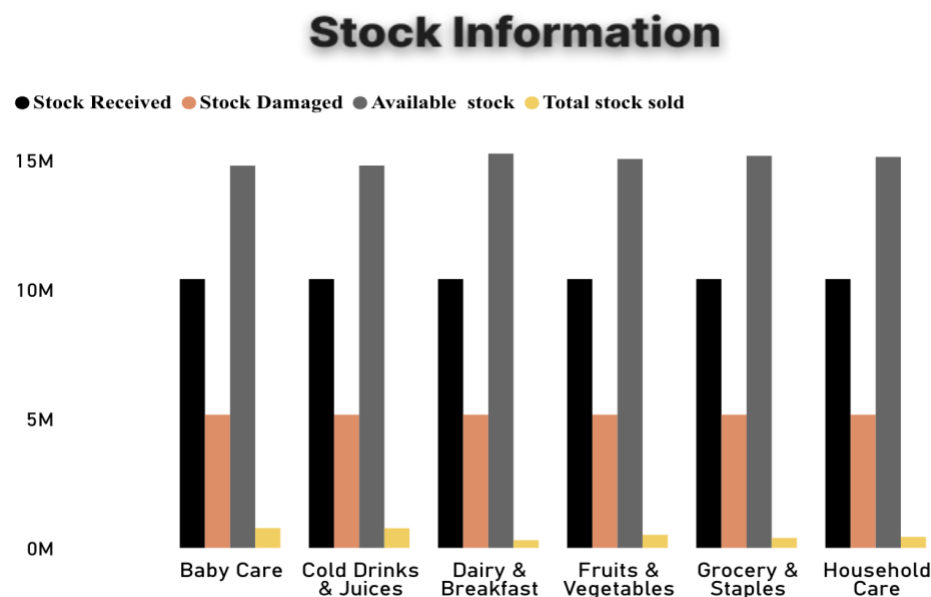


Business Implications:

1. **Balanced Focus Needed:**
 - Since no single feedback category dominates, businesses must maintain a **holistic improvement strategy** across service, product, delivery, and app UX.
2. **Capitalize on Positivity:**
 - With over 70% positive sentiment, the company should **leverage testimonials, reviews, and case studies** to build trust and market credibility.
3. **Address Negative Sentiment Early:**
 - The 9.93% negative sentiment, though small, should not be ignored. **Proactive customer support and issue resolution mechanisms** can reduce this further.
4. **Turn Neutral into Positive:**
 - Investigate neutral feedback to identify **opportunities to delight customers** — even small improvements can increase loyalty and NPS (Net Promoter Score).

Inventory Page

1. Category-wise Stock Analysis (Bar Chart)



Data Shown -:

- High Stock Availability Across Categories:**
All product categories maintain strong levels of available stock, indicating robust supply chain operations and readiness to meet consumer demand.
- Consistent Stock Intake:**
Stock received is fairly uniform across categories, suggesting balanced procurement planning and a streamlined sourcing process.
- Stock Damage Levels:**
Damage levels remain relatively high and consistent across all categories, with **no category falling significantly below the average**. This points to a systemic issue in handling, packaging, or storage rather than category-specific issues.
- Sales Volume (Stock Sold):**
Despite high availability, the **total stock sold remains significantly low across all categories**, as indicated by the short yellow bars. This suggests:
 - Either limited demand or ineffective marketing
 - Possible overstocking, leading to higher holding costs

Business Implications:

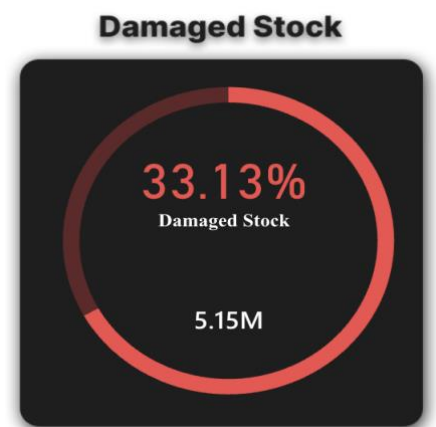
- Optimization Opportunity:**
The gap between available stock and stock sold highlights potential overstocking. This could lead to increased warehousing costs and product expiry risks, especially in perishables like **Dairy & Breakfast** and **Fruits & Vegetables**.
- Need for Strategic Promotions:**
To boost sales and clear stock, targeted promotions, discounts, or bundling strategies should be introduced — especially in slower-moving categories.
- Improve Stock Handling Protocols:**
With uniformly high stock damage across the board, Blinkit must reassess its logistics, packaging, and handling practices to prevent product losses and preserve margins.
- Category Prioritization:**
Categories like **Baby Care**, **Pharmacy**, and **Cold Drinks & Juices** show marginally higher sales, suggesting these could be leveraged further with specialized campaigns or inventory prioritization.

2. Stock Quality Overview (Heat map)

This section presents a visual comparison of the overall stock status using circular gauges, summarizing the proportion of Total Available Stock and Damaged Stock. The data can also be filtered by product category for more granular insights.

Data Shown

-  **Available Stock:**
 - **63.72%** of total stock
 - Representing **9.90 million units**
 - Indicates the proportion of usable, sellable inventory currently in storage.
-  **Damaged Stock:**
 - **33.13%** of total stock
 - Equivalent to **5.15 million units**
 - Represents inventory that cannot be sold due to spoilage, mishandling, or quality issues.



Business Implications:

- **High Damage Rate Signals Operational Risk:**
With over **one-third of the inventory being damaged**, this indicates a **significant efficiency gap** in storage, transportation, or quality control processes.
- **Margin and Cost Impact:**
Damaged stock directly affects profitability, as these items may need to be written off or returned. Reducing the damage rate even by 5–10% could translate into **millions in cost savings**.
- **Inventory Utilization:**
The **63.72% usable stock rate** is relatively healthy but has room for improvement. Maximizing the usable stock percentage ensures better **inventory turnover**, faster sales, and higher return on investment (ROI).
- **Need for Root Cause Analysis:**
A deeper analysis is required to identify **damage patterns** — by product category, warehouse, or supplier — to implement corrective actions.

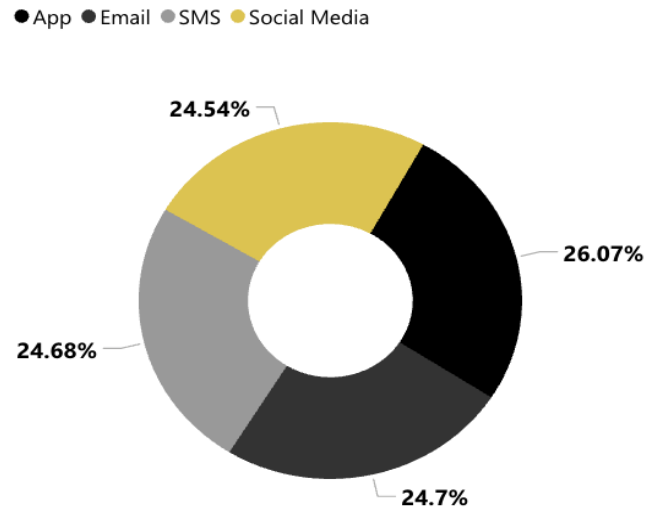
Marketing Page

1. Customer Acquisition by Channel (Pie Chart)

Data Shown -:

- **App** contributed the highest share at **26.07%**, highlighting its effectiveness as a direct acquisition platform.
- **Email** marketing follows closely at **24.7%**, showcasing strong engagement through targeted campaigns.
- **SMS** accounts for **24.68%**, still playing a relevant role in reaching mobile users directly.
- **Social Media** contributes **24.54%**, indicating consistent performance and reach via platforms like Instagram, Facebook, etc.

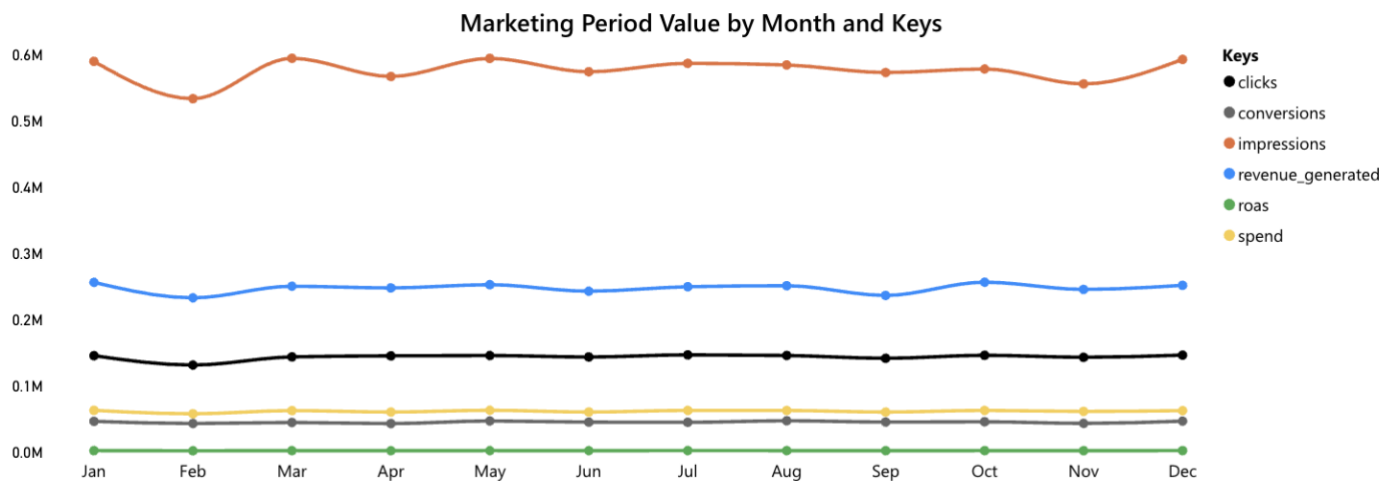
Customer Acquisition by Channel



Business Implications:

- **Balanced Channel Performance:** The near-equal distribution across all four channels indicates a **well-diversified marketing strategy**. No single channel is overly dominant or underperforming.
- **Opportunity to Optimize App Engagement:** Given its slightly higher share, the app presents an opportunity to deeper engagement through personalized features, in-app promotions, and retention strategies.
- **Potential to Scale Social Media:** Despite being slightly lower, Social Media still contributes nearly a quarter of customer acquisitions. With further investment in targeted ads and influencer marketing, it could become a top acquisition driver.
- **Data-Driven Campaign Allocation:** These insights can help allocate marketing budgets and campaign efforts more efficiently based on channel performance.

2. Annual Marketing Performance Summary (Jan–Dec) (Line Chart)



Data Shown -:

- **Impressions** consistently led all metrics, fluctuating slightly between ~**530K and 600K**, with **notable peaks in March, May, and December**, indicating strong visibility during key seasonal pushes or promotions.
- **Revenue Generated** maintained steady performance between **230K–260K**, with **October and December** showing uplift, potentially due to holiday campaigns or end-of-year offers.
- **Clicks** held stable around **130K–150K**, peaking modestly in **March and December**, reflecting consistent user interest.
- **Conversions** (gray line) remained relatively flat, suggesting that while traffic and revenue were stable, **conversion optimization may need improvement**.
- **Spend** hovered around **60K–70K** monthly, reflecting controlled budget management, with minor increases toward the end of the year.
- **ROAS** stayed low and steady, showing **slight improvement in Q4**, which may point to better campaign targeting or seasonal buying behavior.

Business Implication -:

- **Holiday Campaign Efficiency:** October–December saw increases in both revenue and impressions. Consider leveraging this season with enhanced targeting or promotions.
- **Mid-Year Dip:** May to September reflects slight stagnation in performance. Use this time to test new creatives or channels.
- **Conversion Bottleneck:** Despite stable traffic and spend, conversions didn't grow proportionally. Review funnel optimization strategies (e.g., landing pages, CTAs, user experience).
- **Budget Allocation:** Spending is well-distributed, but further ROAS improvements could be achieved by reallocating spend based on channel performance insights.